Linear Regression Analysis Report

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1. Title Page

- **Project Title:** Predicting Insurance Charges Using Linear Regression
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- **Course/Module:** PDAN 8411

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2. Executive Summary

This study examines how personal and lifestyle characteristics influence medical insurance costs. A linear regression model is used to predict charges based on factors such as age, BMI, smoking habits, and number of children. The model performs well for typical cases but is less accurate for extremely high charges.

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3. Introduction

- **Objective:** Identify key factors influencing insurance charges and create a predictive model.

- **Dataset:** 1,338 records from Kaggle, containing demographic, lifestyle, and insurance cost information.

We want to find patterns in the data and make predictions using a simple method called linear regression.

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4. Data Description

- **Numerical features:** age, BMI, children, charges (target)
- **Categorical features:** sex, smoker, region
- **Missing values:** None
- **Observations:** Charges skewed by high values for smokers and older individuals.

Some features are numbers, some are categories, and charges (the cost) is our prediction target.

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5. Methodology

1. **Data preprocessing:** One-hot encoding for categorical variables; split into 80% train and 20% test sets.
2. **Model training:** Linear regression fitted on training data.
3. **Evaluation metrics:**
    - MAE – mean average error
    - RMSE – average prediction error
    - R^2 – proportion of variance explained

We converted text to numbers, trained the model, and checked how accurate it is.

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6. Results & Analysis

- Predicted vs actual charges align closely for most data points.
- Outliers exist for extremely high charges (mainly older smokers).
- Residuals roughly normal → model assumptions mostly valid.

The model works well for typical cases but struggles with very high costs.

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7. Key Findings

1. **Significant factors:** smoker status, age, BMI
2. **Model effectiveness:** strong for average charges; less precise for extreme high charges
3. **Business insight:** consider high-risk profiles in insurance pricing
4. **Recommendations:** Consider using polynomial regression or applying feature transformations to enhance the model's accuracy for outlier predictions.

Smoking, age, and BMI are most important. The model is useful, but further improvement is possible for extreme cases.

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8. Conclusion

Linear regression provides a simple yet effective method for predicting insurance charges. While it captures the general trends, handling extreme values may require more advanced techniques. The analysis demonstrates the ability to interpret data and make informed decisions based on predictive modelling.

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9. References

Kaggle, 2026. *Medical Cost Personal Dataset*. [online] Available at: https://www.kaggle.com/datasets/mirichoi0218/insurance [Accessed 5 April 2026].

Landbot, 2026. *Landbot Online Chat Flow*. [online] Available at: https://landbot.online/v3/H-1471321-8E8ERBM520BXNWZV/index.html [Accessed 5 April 2026].

YouTube, 2026a. *Linear Regression Tutorial*. [video] Available at: https://www.youtube.com/watch?v=sV5JUFFResA [Accessed 6 April 2026].

YouTube, 2026b. *Polynomial Regression Explained*. [video] Available at: https://www.youtube.com/watch?v=yYvSscuM8so [Accessed 7 April 2026].

Carnegie Mellon University, 2026. *Paper Structure Notes*. [online] Available at: https://www.stat.cmu.edu/~brian/701/notes/paper-structure.pdf [Accessed 9 April 2026].